

# Paper III — Final Internal Audit Report

Linear-Error Clique Partitions of Split Graphs (Erdos #81)

**OVERALL VERDICT: ALL AUDITED BLOCKS PASS**

## 1. Scope

This report consolidates the independent internal audit of the finite and closed-form claims of Paper III. Every claim was re-derived from scratch, independently of the manuscript's own scripts and of the Lean 4 / Mathlib formalization. Numeric grid checks use exact rational arithmetic; algebraic identities are proved symbolically with SymPy; fractional optima use direct linear programming (HiGHS); integral triangle-packing numbers use exact 0/1 integer programming (CBC). The two external asymptotic inputs AX1 (Haxell-Rodl/Yuster) and AX2 (Dross + Barber-Kuhn-Lo-Osthus) are the paper's declared axioms and are outside the scope of computational audit.

## 2. Results by block

### Block 01 — Algebraic identities

**VERDICT: 12/12 identities PASS**

Method: SymPy exact symbolic proof ( $\text{simplify}(\text{LHS}-\text{RHS})=0$  / exact rational / sum-of-squares) of the  $T(G)$  key identity (Thm 4.2), the (9.12) and (9.20) coefficients, the (9.19) completed square and its lower bound,  $\delta \geq 7/8$  for both parities (9.10), the corridor threshold  $p=2304$ ,  $\mu$  continuity at  $\alpha=2/3$ , and the (4.5) closed forms.

results SHA-256: 7d5ea305ff7f44013184d2187d7e310efde926881cf24a4424e55667dbfe8519

zip SHA-256: f0bf84900c54f70e43d9c3bc77aab8010d51b034924b2c46724fb961c0584d15

### Block 02 — Common-profile LP (Theorem 3.1 / E-3.1)

**VERDICT: LP 351/351 and EXACT 351/351 PASS**

Method: Direct fractional triangle-packing LP (SciPy HiGHS) on the actual graph  $H(p,q,d)$  over  $3 \leq p \leq 8$ ,  $0 \leq q \leq 8$ ,  $0 \leq d \leq p$ , compared to the closed form  $F(p,q,d)$ ; PLUS an exact-rational feasible-cover certificate proving  $\nu_3^* \leq F$  without float reliance.

results SHA-256: 1e977ecba3d6475adf02330f8fb6d0c382d6123d83ead0bb468c7287c3e79773

zip SHA-256: 2b44615bcca18af350c5b0328a598c4deeb28d877170fe8e1fa0f0746b80b68e

### Block 03 — Unified fractional margin (Theorem 4.2 / E-4.2)

**VERDICT: 78,384/78,384 exact-rational checks PASS**

Method: Exact-rational grid audit ( $\text{fractions.Fraction}$ , no floating point) of the completion-of-squares inequality (4.5) over  $3 \leq p \leq 48$ ,  $1 \leq q \leq 2p$ ,  $0 \leq d \leq p$ , with third-branch dominance bookkeeping.

results SHA-256: 577450739c3652ed9be447a3b426f23f8e856dd7eac4aa37002bcd859b5239a

zip SHA-256: f42c326551e0254b4e9d98581759c2062951e1af4275b6972d482189e83cfa32

### Block 04 — Corridor integral packing (Lemma 5.1 & Cor 5.3)

**VERDICT: E-5.1 180/180, Cor 5.3 180/180, 372 instances PASS**

Method: Exact 0/1 ILP (PuLP + CBC) computation of  $\nu_3(G)$  on 372 systematically generated split graphs, verifying E-5.1 and Corollary 5.3 on the applicable instances ( $q \geq r_p$ ) plus the basic invariants  $0 \leq \Phi$  and  $3 \cdot \nu_3 \leq |E|$ .

results SHA-256: eeb55cbb2b33abcbeae841b6ed2562b1a3ba039c3dfdbe5f0074d2561b9fa7ce

zip SHA-256: 7461f221ed91623fd2f04ccee4baa4fc013211b1468fb03a1b7fcbf428009aa7

### 3. Methodology & tooling

Python 3.14; SymPy 1.14 (symbolic identities); SciPy 1.17 HiGHS (fractional LP); PuLP + CBC (exact ILP); fractions.Fraction (exact rationals). Each block's `verify_*.py` writes its full log to `results/` and returns nonzero on any failure. The per-block PDF certificates and zips are produced by `build_audit_artifacts.py`; this master report is produced by `make_master_certificate.py`.

### 4. Relationship to the formalization

This computational audit is complementary to the machine-checked Lean 4 / Mathlib development. The formalization certifies the full logical chain (Theorem 1.1 and the elementary core) relative to AX1 and AX2; this audit independently certifies the finite numeric and closed-form facts those proofs rely on. Agreement between the two is corroborating, not circular: they share no code.

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Auditor: internal automated audit harness (Claude Code)

*This certificate attests to computational and symbolic verification of the stated finite/closed-form claims at the audited parameter ranges.*